ROUND 3: THE FINISH — PRODUCT RELEASE

Solo Hackathon Track • Inkloom AI Brand-Governance

InkSentinel AI

AI-Assisted Enterprise Brand Governance, Visual WCAG Guard & Cryptographic Audit Platform

LEAD ENGINEER & AI ARCHITECT
Team Sentinel (Solo) •
24eg105f30@anurag.edu.in

LIVE PRODUCTION DEPLOYMENT
inksentinel-brand-governance.vercel.app

PUBLIC SOURCE CODE REPOSITORY
github.com/24eg105f30-lgtm/inksentinel-brand-governance

JUDGE SELF-GUIDED EVALUATION
Credentials: sarah.copywriter / marcus.brandlead

01 Product Overview & Complete Feature List

25 Points

InkSentinel AI is an assistive brand governance co-pilot designed to resolve the tension between rapid content production and regulatory exposure. Unlike blunt black-box LLM wrappers, InkSentinel performs deterministic, explainable audits against Brand Bible clauses (§1.3, §2.1, §3.4, §4.2, §4.5, §5.1) and Approved Fact Bases (#SEC-104, #PRC-201, #FEAT-305, #PERF-402). Suggestions are strictly advisory; no copy is published or silently altered without creator approval.

Complete Feature List (Round 3 Hardened Release)

1. **Text & Claim Studio:** Split-pane workspace with real-time Brand Health Score (0–100) across 5 core violation categories.
2. **Explainable Redline Diff:** Character-span diffs citing clause IDs and grounded fact records with 1-click adoption.
3. **Token-Level Redline Canvas:** Inline word-level diff visualizer with editable replacement tokens before adoption.
4. **Hybrid Claim Grounding Sandbox:** Tri-state claim verification (`VERIFIED`, `CONTRADICTED`, `UNVERIFIED`) enforcing the zero-invented-facts invariant.
5. **Client-Side Regex PII Masking:** Pre-inference redaction of emails, phones, names, and API keys before external transit.
6. **Visual & WCAG Guard:** Canvas luminance calculator evaluating text against background pixels with interactive region selection.
7. **Brand Palette Conformity Guard:** Hex distance validator checking colors against 5 approved enterprise brand tokens.
8. **AI Tone Review (Gemini Flash):** Advisory tone review via Vercel serverless function with deterministic edge fallback.
9. **Role Enforcement & Comments:** Creator vs Reviewer role toggles with inline threaded comments saved to localStorage.
10. **Cryptographic Audit Sign-Off:** Web Crypto SHA-256 certificate generation with exportable JSON audit trail receipt.
11. **Guideline Ambiguity Heatmap:** Infraction analytics with simulated telemetry and AI-assisted policy rewrite suggestions.
12. **Fact Base Ingestion Modal:** Runtime GUI dialog allowing compliance teams to ingest new verified fact records.
13. **Local Python REST Server & Test Suite:** Mirror server (`server.py`) and unit test suite (`test_suite.py`) with zero pip dependencies.

02 The Three Named Improvements Delivered (Committed at Round 2)

20 Points

Named Impr. #1 Deepened Visual & WCAG Guard

Round 2 Commitment: OCR finds text blocks, measures contrast against background pixels, marks failing regions, checks palette.

Delivered: HTML5 Canvas luminance analyzer. Uses preset bounding boxes on demo ad banners to sample foreground and background pixels directly behind text blocks, computing WCAG 2.1 relative luminance `(L1+0.05)/(L2+0.05)`. Highlights passing (green) vs failing (red) regions. Color distance engine checks conformity against 5 brand palette tokens (#1E1B4B, #4338CA, #0F172A, #4A5568, #F8FAFC).

Named Impr. #2 Grounded Fact & Claim Verification

Round 2 Commitment: Hybrid keyword & embedding retrieval, claims labeled verified/contradicted/unverified with citations, PII masking.

Delivered: Hybrid retrieval combining token overlap with character n-gram cosine vector similarity against `fact_base.json`. Tri-state labeling: Verified (#SEC-104), Contradicted (prohibited claims), and Unverified (hallucinations blocked). Client-side regex PII engine scrubs emails, phones, names, and API keys.

Named Impr. #3 Multi-Role Review & Redline Workflow

Round 2 Commitment: Token-level editable redline, enforced Creator/Reviewer sign-in, saved drafts, reviewer comments, stored sign-off.

Delivered: Token-level inline-editable redlines with interactive replacement tokens. Enforced Creator (Sarah) and Reviewer (Marcus) roles. Threaded reviewer commenting per finding. Multi-slot local draft persistence. Web Crypto SHA-256 certificate generation with persistent audit history.

03 Technical Architecture & Truthful Implementation Summary

Architecture

Technology Stack & Network Boundaries

- **Browser Client:** Single-Page Architecture using semantic HTML5, Tailwind CSS, Lucide Icons, and HTML5 Canvas. Core deterministic rule audits, WCAG luminance math, regex PII masking, and Web Crypto SHA-256 hashing run 100% locally on-device. CDN assets (Tailwind, icons) require initial internet connectivity.
- **Edge Serverless AI (Vercel):** `api/tone.js` calls Google Gemini (gemini-1.5-flash) via an OpenAI-compatible endpoint. Features dual-layer fallback heuristics: if the key is missing or rate-limited, local brand heuristics score the copy with zero crashes.
- **Grounding & Retrieval:** Implemented as hybrid keyword plus character n-gram cosine vector similarity against JSON facts—a lightweight, dependency-free stand-in for heavy vector DBs.

Local Python REST Server & Algorithms

- **Standard Library Backend (`server.py`):** Uses standard Python 3.9+ (zero pip packages). Serves static files and REST endpoints: `/api/health`, `/api/guidelines`, `/api/facts`, `/api/ambiguity`, `/api/wcag`, and `/api/audit`.
- **WCAG Luminance Formula:** Deterministic sRGB gamma correction (ISO/IEC 40500:2012): `L = 0.2126R' + 0.7152G' + 0.0722B'`.
- **Cryptographic Audit Payload:** Generates tamper-evident SHA-256 digest hashing: `${text}|${author}|${reviewer}|${timestamp}|BrandBible-2.4.0`.

04 Reliability Under Normal Use & Testing Details

20 Points

In accordance with Round 3 judging criteria ("Make it hold up without you: bad input, an empty state, a refresh, the same action twice — nothing should break or hang"), InkSentinel was stress-tested across edge cases. Core rule checks run natively in the browser with zero external API dependencies required for base governance.

Automated Test Suite (test_suite.py)
Result: `Ran 7 tests in 0.006s — OK (7 of 7 passed)`
Unit tests validate Python server logic and core deterministic algorithms:
- `test_wcag_contrast_failing`: Confirms 2.26:1 contrast failure on #A0AEC0/white.
- `test_wcag_contrast_passing`: Confirms 7.53:1 contrast pass on #4A5568/white.
- `test_pii_masking_scrubs`: Verifies regex redaction of emails, phones, names.
- `test_claim_contradiction`: Flags absolute claims vs #SEC-104.
- `test_claim_verification`: Confirms verified assertion matching.
- `test_claim_unverified`: Rejects ungrounded marketing claims.
- `test_cryptographic_signoff`: Asserts 64-char SHA-256 certificate generation.

Manual End-to-End Test Matrix
- **Empty Text:** Shows "0 Infractions Flagged" & clean state; no crashes.
- **Very Long Copy:** 2,000+ words parsed instantaneously (<20ms).
- **Accept Twice:** First click adopts fixes; second reports 0 violations.
- **Refresh Mid-Journey:** Draft comments & sign-off history persist via localStorage.
- **Bad Custom Upload:** Displays demo scope notice and renders gracefully.
- **Sign-off by Creator vs Reviewer:** Creator button is disabled; Reviewer enables full sign-off.
- **Missing AI Key / Outage:** Dual-layer fallback catches errors; tone review scores smoothly.

05 Usability, Real-World Fit & Exact Judge Walkthrough

20 Points

InkSentinel AI bridges editorial copywriters and brand safety officers. Role enforcement is implemented as client-side UI state toggles with localStorage persistence (no server-side authentication), allowing judges to immediately evaluate cross-functional workflows.

Role Profile	Demo Credentials	Permitted Capabilities	Recommended Judge Journey (Exact Labels)
Content Creator (Sarah Jenkins)	`sarah.copywriter@inkloom.ai` Password: `sentinel-creator-2026`	Drafting, audit execution, 1-click diff adoption, PII inspection, draft saving.	1. Select `1. SaaS launch copy` from **Load example…** 2. Click **Run Audit**, review flagged findings. 3. Click **1-Click Accept All Diffs**; score rises from 60% to 100%.
Brand Director (Marcus Vance)	`marcus.brandlead@inkloom.ai` Password: `sentinel-lead-2026`	Reviewer comments, claim verification, cryptographic sign-off authorization.	1. Switch top toggle to **Marcus (Reviewer)**. 2. Add a reviewer note on finding #SEC-GOV-04. 3. Switch to **Reviewer Sign-Off & Audit** tab. 4. Click **Official Brand Sign-Off** & **Download Certificate (JSON)**.

06 Known Limitations & Commercial Roadmap

15 Points

Declared Scope Boundaries & Limitations
- **Role Enforcement:** UI-level demo toggles with localStorage; no production OAuth/SSO.
- **OCR Scope:** Uses preset bounding coordinates for the built-in ad banner. Custom image upload demonstrates the workflow and falls back to preset demo bounding regions; arbitrary in-browser OCR is on the roadmap.
- **Ambiguity Heatmap:** Uses simulated historical baseline telemetry (1,420 simulated audits) + live telemetry.
- **Persistence:** LocalStorage is per-device/browser and clears on cache wipe.

Commercial Development Roadmap (Post-Hackathon)
- **Q1 2027:** Native Figma plugin and Adobe Creative Cloud extension for visual WCAG inspection on canvas; integration of in-browser WebAssembly OCR (Tesseract.js).
- **Q2 2027:** CI/CD GitHub Action and Headless CMS (Contentful/Sanity) webhooks blocking non-compliant copy before deployment.
- **Q3 2027:** Enterprise SSO (Okta/SAML 2.0), multi-tenant Fact Base isolation, and SOC-2 audit logs.
- **Q4 2027:** Localization support for global Brand Bibles with localized regulatory rules.

Verification & Submission Links: Live Vercel App: https://inksentinel-brand-governance.vercel.app • GitHub Repository: https://github.com/24eg105f30-lgtm/inksentinel-brand-governance • Solo Lead: Team Sentinel (24eg105f30@anurag.edu.in).